

Air Quality Index (AQI) Classification Using Hybrid Multi-Input CNN

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Abstract—Air pollution has become one of the most critical environmental and public health challenges faced by modern urban societies. Rapid urbanization, industrial growth, and the increasing number of vehicles have significantly contributed to the deterioration of air quality in many cities around the world. Poor air quality not only harms the environment but also poses serious health risks to humans, including respiratory diseases, cardiovascular problems, and other long-term health complications. To monitor and communicate the severity of air pollution, governments and environmental agencies use the Air Quality Index (AQI), a standardized scale that translates complex air pollution data into an easily understandable format for the public.

Traditional AQI monitoring systems rely heavily on physical air quality sensors that measure pollutants such as particulate matter (PM_{2.5} and PM₁₀), nitrogen dioxide, sulfur dioxide, and carbon monoxide. While these sensors provide accurate measurements, they are expensive to install and maintain. Additionally, because of their high cost and infrastructure requirements, they are usually deployed only at selected monitoring stations, creating limited spatial coverage, particularly in developing countries.

This paper proposes a novel approach to estimating air pollution levels by leveraging computer vision and deep learning techniques. The proposed method uses environmental images to predict the Air Quality Index using a Hybrid Multi-Input Convolutional Neural Network (CNN) architecture. The model integrates three representations of the same image: the original RGB image, saturation-based images, and superpixel-segmented images. Using EfficientNetB0 as the backbone with transfer learning, the proposed architecture achieves a validation accuracy of approximately 98.67%, an F1 score of 0.986 across all six AQI classes, and a classification loss of 0.043 on the validation set, demonstrating the effectiveness of combining multiple image representations for scalable, cost-effective AQI estimation.

Index Terms—Air Quality Index, Convolutional Neural Network, EfficientNetB0, Transfer Learning, Superpixel Segmentation, Grad-CAM, Environmental Monitoring, Deep Learning.

I. INTRODUCTION

Air pollution has emerged as one of the most pressing global environmental challenges of the modern era. According to the World Health Organization (WHO), approximately 99% of the global population breathes air that exceeds WHO guideline limits, and ambient air pollution is responsible for an estimated 4.2 million premature deaths per year worldwide. It not only affects the quality of the air we breathe but also has far-reaching consequences for public health, climate stability, and the overall sustainability of urban environments. Over the past

few decades, rapid industrialization, increased vehicular emissions, population growth, and uncontrolled urban expansion have significantly contributed to worsening air quality in many parts of the world. Major cities, particularly in developing countries, frequently experience dangerous pollution levels that can negatively impact millions of people on a daily basis. India alone accounts for 21 of the world's 30 most polluted cities as per the IQAir 2023 World Air Quality Report. Long-term exposure to polluted air has been linked to respiratory diseases, cardiovascular problems, reduced lung function, and other severe health complications. Therefore, monitoring air quality has become a critical task for governments, researchers, and environmental organizations in order to protect public health and develop effective environmental policies.

One of the most widely used tools for communicating air pollution levels to the public is the Air Quality Index (AQI). The AQI serves as a standardized indicator that converts complex air pollution measurements into a simplified format that is easy to interpret. Instead of presenting raw pollutant concentration values, the AQI categorizes air quality into different levels such as *Good* (0–50), *Moderate* (51–100), *Unhealthy for Sensitive Groups* (101–150), *Unhealthy* (151–200), *Very Unhealthy* (201–300), and *Severe* (301–500). Each category represents a specific range of pollution levels and provides guidance on potential health effects, enabling citizens to quickly understand the quality of the air around them and make informed decisions about outdoor activities and safety precautions.

Traditionally, AQI measurements are obtained using specialized air quality monitoring stations equipped with physical sensors. These monitoring stations are capable of providing accurate and reliable data; however, installing and maintaining such infrastructure is often expensive and technically demanding, with a single monitoring station costing between \$15,000 and \$40,000 USD to set up and approximately \$5,000 per year to maintain. As a result, these monitoring systems are typically deployed only in selected locations within major cities, leading to significant gaps in spatial coverage. India, for instance, operates only around 800 CPCB-approved monitoring stations across the entire country, translating to approximately one station per 4,100 km².

In recent years, advances in deep learning and computer vision have opened new possibilities for addressing these

limitations. Environmental images captured by cameras, smartphones, or surveillance systems often contain valuable visual information about atmospheric conditions. Features such as haze intensity, reduced visibility, sky color variations, and atmospheric scattering are closely associated with pollution levels in the air. By analyzing these visual cues using machine learning models, it becomes possible to estimate AQI levels directly from images. This emerging approach has the potential to significantly enhance air quality monitoring capabilities by leveraging widely available camera networks and mobile devices to collect environmental data across larger geographical areas at a fraction of the cost of deploying physical sensors.

II. PROBLEM STATEMENT AND OBJECTIVES

Despite significant advancements in air quality monitoring technologies, traditional sensor-based AQI measurement systems still face several practical limitations. Most conventional monitoring methods depend on specialized sensor stations that measure pollutant concentrations in the atmosphere. Although these systems provide accurate and reliable data, they are expensive to install, operate, and maintain. Setting up a large network of monitoring stations requires substantial financial investment, technical infrastructure, and continuous calibration. As a result, many cities and regions have only a limited number of monitoring stations, leading to restricted geographical coverage. In India, the Central Pollution Control Board (CPCB) maintains around 800 real-time stations across 344 cities, leaving a majority of the country's 640,000+ villages and numerous smaller towns without any reliable AQI monitoring.

Another challenge arises when attempting to analyze environmental conditions using visual observations alone. While humans can often notice signs of pollution such as haze, reduced visibility, or changes in sky color, manually estimating pollution levels from images is highly subjective and inconsistent. Environmental images are influenced by various external factors such as lighting conditions, time of day, weather variations, camera quality, and scene complexity, making accurate and consistent classification even more challenging through manual methods.

To address these challenges, this project aims to develop a robust and intelligent deep learning model capable of automatically classifying Air Quality Index (AQI) levels using environmental images. The primary objectives of this study are:

- 1) Design and implement an effective hybrid multi-input CNN architecture capable of learning complex environmental features from RGB, saturation, and superpixel image representations.
- 2) Enhance classification robustness by incorporating feature-enhanced image inputs that highlight important atmospheric characteristics invisible to a single-stream model.
- 3) Achieve high model performance, targeting a validation accuracy above 95% across all six AQI categories during training and evaluation.

- 4) Incorporate explainable AI techniques, particularly Grad-CAM (Gradient-weighted Class Activation Mapping), to provide interpretable and trustworthy predictions suitable for institutional deployment.
- 5) Deploy the trained model as an accessible web application that allows end users to submit images and receive real-time AQI predictions.

Through these objectives, the project aims to contribute toward developing a more scalable, intelligent, and interpretable solution for air quality monitoring using modern deep learning technologies.

III. LITERATURE REVIEW

Over the past decade, the rapid advancement of deep learning and computer vision has encouraged researchers to explore new approaches for estimating air pollution using visual data. Several studies have investigated the potential of image-based air pollution estimation as a complementary alternative to traditional sensor-based monitoring systems.

Early research demonstrated that convolutional neural networks (CNNs) are highly effective for extracting meaningful visual features from environmental images [2]. CNNs automatically learn hierarchical representations of image data, allowing them to detect patterns such as haze density, sky brightness, and atmospheric clarity without requiring manual feature engineering.

In addition to purely deep learning-based approaches, several studies have combined CNN models with traditional image processing techniques to improve performance. Classical methods such as the Dark Channel Prior (DCP) and atmospheric scattering models have been widely used in image dehazing research [1]. The dark channel prior technique is particularly useful for identifying haze concentration in images. When incorporated into deep learning frameworks, these techniques help models better capture haze-related information that may be difficult to detect using standard RGB images alone.

More recent research has focused on hybrid CNN architectures that combine handcrafted features with automatically learned deep features [3]. Handcrafted features such as dark channel maps can explicitly highlight haze patterns, while saturation channels emphasize color intensity variations caused by suspended particulate matter in the atmosphere. Another important development is the use of multi-stream or multi-input CNN architectures, in which different representations of the same image are processed through separate neural network branches before being combined [5].

Furthermore, recent studies have explored lightweight deep learning architectures and autoencoder-based models for efficient pollution estimation [3]. Some researchers have also incorporated attention mechanisms into CNN architectures to help the model focus on the most informative regions of an image [5]. The collective findings from these previous studies highlight the importance of combining multiple sources of visual information when estimating air pollution from images.

Table I summarizes the key related works and their methodological contributions.

IV. DATASET DESCRIPTION

The model used in this study was trained using the Air Pollution Image Dataset, which consists of environmental images collected from various regions in India and Nepal. These regions often experience varying levels of air pollution due to factors such as industrial activities, vehicular emissions, seasonal crop burning, and changing weather conditions. As a result, the dataset provides a diverse range of atmospheric scenarios useful for training and evaluating deep learning models. The images capture real-world environmental scenes, including urban landscapes, roads, buildings, and open skies, where visible pollution indicators such as haze, fog, and reduced visibility are present.

The dataset consists of approximately 7,200 labeled images distributed across six AQI categories. The class-wise distribution prior to augmentation is shown in Table II. Images are sourced from publicly available environmental image repositories, webcam snapshots, and manually curated photo collections from Indian and Nepalese cities.

To facilitate effective classification, the dataset categorizes images into the six distinct AQI levels described above. This multi-class categorization enables the model to learn nuanced differences between various pollution levels rather than simply distinguishing between clean and polluted air.

A. Preprocessing

Before training the neural network, several preprocessing steps were applied to ensure consistency and compatibility with the deep learning architecture. Each image in the dataset was resized to a standardized resolution of 224×224 pixels. This resolution was selected because it matches the input size requirements of EfficientNetB0 when employing transfer learning. Pixel values were normalized to the range $[0, 1]$ by dividing by 255, and subsequently standardized using ImageNet mean ($\mu = [0.485, 0.456, 0.406]$) and standard deviation ($\sigma = [0.229, 0.224, 0.225]$) channel-wise, which helps stabilize training and improves convergence during optimization.

B. Data Augmentation

Data augmentation techniques were incorporated to enhance dataset diversity and improve the model's ability to generalize to unseen images. The following augmentation operations were applied on-the-fly during training:

- Random rotation within $\pm 20^\circ$
- Horizontal flipping with a probability of 0.5
- Random zoom scaling between $0.85\text{--}1.15\times$
- Random brightness adjustment ($\pm 15\%$)
- Random contrast jittering ($\pm 10\%$)

These operations simulate variations that might naturally occur in real-world images, such as changes in camera orientation, perspective, lighting conditions, or scene composition.

C. Class Imbalance Handling

One of the key challenges encountered was class imbalance within the dataset. The *Good* category contained the most samples (1,450), while *Very Unhealthy* contained the fewest (940) — a ratio of approximately 1.54:1. Although relatively moderate compared to typical imbalanced datasets, this discrepancy was sufficient to bias the model. To address this, a class weighting strategy was implemented during the training phase using inverse-frequency weights. Specifically, the weight for class c was computed as:

$$w_c = \frac{N}{K \cdot n_c} \quad (1)$$

where N is the total number of samples, K is the number of classes, and n_c is the number of samples in class c . By assigning higher weights to minority classes, the training process encourages the model to pay greater attention to underrepresented categories, resulting in more balanced per-class F1 scores.

V. IMPACT OF DATASET ON MODEL PERFORMANCE

The quality, diversity, and scale of the training dataset play a central role in determining the generalization capability of any deep learning model. In the context of image-based AQI classification, the dataset used for training not only determines what visual patterns the model learns but also governs how reliably those patterns transfer to real-world deployment scenarios.

A. Dataset Diversity and Geographic Bias

The dataset used in this study was sourced primarily from Indian and Nepalese cities and towns, which introduces a geographic bias into the trained model. The model learns visual pollution signatures specific to the environmental and infrastructural characteristics of the Indian subcontinent — including typical sky colors, urban building materials, vegetation density, and dust characteristics common to northern Indian plains. While this is beneficial for deployment within the same geographic region, it may limit the model's applicability in cities with substantially different visual environments such as coastal cities, desert regions, or heavily forested urban areas.

Furthermore, the dataset predominantly captures daytime outdoor scenes. The model has not been trained on nighttime images or indoor environments where pollution can still be significant but is not visually manifested through haze or reduced visibility in the same way. This limits the operational scope of the current system.

B. Label Quality and AQI Ground Truth

A critical consideration is the reliability of ground-truth AQI labels. Since images cannot independently carry embedded sensor readings, the AQI labels assigned to images in the dataset were derived by correlating image capture timestamps and locations with nearby CPCB monitoring station readings. This introduces an inherent labeling uncertainty: the actual AQI at the exact image location at the exact moment of capture

TABLE I
SUMMARY OF RELATED LITERATURE

Paper Title	Authors	Methodology	Key Findings
Image-based Air Pollution Estimation Using Hybrid CNN (2019)	Jian Ma, Kun Li, Yahong Han	Hybrid Deep CNN + Dark Channel Features + Feature Fusion	Dark channel captures haze information. Multi-stream fusion improves robustness.
Image-based Air Quality Analysis Using Deep CNN (2020)	Yilmaz Chakma, Ben Vizona, Tingting Cao, Jerry Lin, Jing Zhang	Deep CNN + Transfer Learning	CNN learns features automatically. Transfer learning improves accuracy. No handcrafted features needed.
Image-Based PM Pollution Analysis Using Lightweight Autoencoder (2021)	Xulong Zhang, Deepak Shrestha, Afsana Khan, Jing Zhang	Lightweight Autoencoder + Saturation Channel + Superpixels	Saturation is a useful pollution indicator. Superpixels preserve structure. Efficient lightweight design.
Multispectral Satellite Image Dehazing & Pollution Estimation (2018–2019)	Sai Avinash G, Shaistha M, Nishen Ganegoda, John Basha M., Naga Kushal Ageeru	Dark Channel Prior + Atmospheric Scattering Model	Haze & pollution strongly correlated. Transmission maps improve estimation.
Two-Stream Non-Uniform Concentration Reasoning Network (2021–2022)	Huilin Chen, Wenming Yang, Qingmin Liao	Two-Stream CNN + Attention Mechanism	Multi-stream improves feature capture. Attention enhances performance.

TABLE II
DATASET CLASS DISTRIBUTION

AQI Category	AQI Range	Image Count
Good	0–50	1,450
Moderate	51–100	1,320
Unhealthy for Sensitive Groups	101–150	1,180
Unhealthy	151–200	1,060
Very Unhealthy	201–300	940
Severe	301–500	1,250
Total		7,200

may differ from the nearest station reading by as much as 20–30 AQI units, depending on distance from the station and local pollution point sources. Such label noise is most problematic for boundary cases — for example, an image captured when AQI was 98 (Moderate) versus 104 (Unhealthy for Sensitive Groups) may be visually nearly identical, yet carry different class labels.

This label uncertainty is reflected in the model’s confusion matrix, where the majority of misclassifications occur between adjacent AQI categories rather than between widely separated ones. Improving labeling precision through denser sensor networks or co-located sensor-camera rigs would directly translate to higher inter-class separability and improved model accuracy.

C. Dataset Scale and Overfitting Risk

With approximately 7,200 images split across six classes, the per-class training set averages roughly 800–1,000 samples before augmentation. This relatively modest scale necessitates the use of transfer learning to avoid overfitting, as a network

trained from scratch on such a dataset would likely memorize training samples. The decision to use EfficientNetB0 pre-trained on ImageNet was directly driven by this constraint. Transfer learning effectively multiplies the useful information available to the model by allowing it to leverage features learned from over 14 million ImageNet images, enabling good generalization even from a small domain-specific training set.

In summary, while the current dataset is sufficient for demonstrating proof-of-concept performance, scaling up to a geographically diverse, multi-season, and temporally rich dataset of 50,000+ images across multiple countries would be the single most impactful improvement for real-world deployment, as discussed further in Section XI.

VI. METHODOLOGY

The proposed system is built around a Hybrid Multi-Input CNN architecture specifically designed to capture complementary information from multiple representations of environmental images.

A. Input Representations

The system generates three distinct image representations to provide the network with complementary atmospheric information.

1) *RGB Image*: The original RGB image preserves the natural visual appearance of the environment, serving as the primary input stream and capturing the overall atmospheric scene. Each pixel is represented as a 3-channel vector $(R, G, B) \in [0, 255]^3$, and the input tensor shape fed to the network is $224 \times 224 \times 3$.

2) *Saturation Image*: The saturation representation is derived by converting the RGB image to the HSV (Hue-Saturation-Value) color space and extracting the saturation

channel. The saturation value S quantifies the colorfulness of each pixel:

$$S = \frac{\max(R, G, B) - \min(R, G, B)}{\max(R, G, B)} \quad (2)$$

Atmospheric pollution often reduces the apparent saturation of environmental scenes because suspended particulate matter scatters light and washes out color. Higher AQI levels typically produce scenes with lower mean saturation. By emphasizing color intensity and saturation levels, this representation helps the model detect subtle visual patterns that may correspond to different levels of air pollution. The saturation image is replicated across three channels to match the EfficientNetB0 input dimensions, yielding a $224 \times 224 \times 3$ tensor.

3) *Supapixel Image*: The superpixel segmentation is computed using the Simple Linear Iterative Clustering (SLIC) algorithm [3], which groups spatially proximate pixels sharing similar color and texture properties into compact segments. In this study, images are segmented into approximately 200–400 superpixels, with each superpixel region assigned the mean color of its constituent pixels. Superpixel images allow the neural network to focus on broader atmospheric patterns such as haze distribution and visibility degradation across different parts of the image, while reducing high-frequency noise and fine texture detail that is irrelevant to AQI classification.

B. Network Architecture

To further enhance the model's performance, transfer learning is employed using EfficientNetB0 as the backbone architecture [6]. EfficientNetB0 is a powerful and computationally efficient convolutional neural network that has been pre-trained on large-scale image datasets. It introduces a compound scaling method that uniformly scales network depth, width, and resolution, achieving an ImageNet top-1 accuracy of 77.1% with only 5.3 million parameters. Each input representation is passed through separate EfficientNetB0 feature extraction branches, allowing the model to learn distinct feature representations for each image type.

After feature extraction, the outputs from the three input streams are combined through feature concatenation, creating a unified feature representation of dimensionality $3 \times 1,280 = 3,840$ (since each EfficientNetB0 branch produces a 1,280-dimensional global average pooled feature vector). The fused feature vector is then passed through several fully connected layers that perform the final classification task. To improve training stability and prevent overfitting, batch normalization and dropout regularization (with $p = 0.4$) are applied within these layers. Finally, the output layer predicts the corresponding AQI category using a softmax activation function with 6 output neurons.

The overall architecture can be summarized as:

$$\mathbf{f}_{fused} = \text{Concat}(f_{RGB}, f_{sat}, f_{sp}) \quad (3)$$

$$\hat{y} = \text{softmax}(W \cdot \text{Dropout}(\text{BN}(\mathbf{f}_{fused})) + b) \quad (4)$$

where f_{RGB} , f_{sat} , and f_{sp} are the feature vectors extracted from the RGB, saturation, and superpixel branches respectively, W is the weight matrix of the classification head, and b is the bias term.

C. Training Configuration

The model was trained using the Adam optimizer with an initial learning rate of 1×10^{-4} for the classification head. During the fine-tuning phase, the last 30 layers of each EfficientNetB0 branch were unfrozen and trained with a reduced learning rate of 5×10^{-6} to preserve pretrained feature representations. Training was conducted over 40 epochs with a batch size of 32, using categorical cross-entropy as the loss function with class weights applied. A learning rate scheduler with cosine annealing was used to gradually reduce the learning rate during the final 10 epochs.

VII. MODEL IMPROVEMENTS

During the early stages of experimentation, the initial version of the model achieved a validation accuracy of approximately 52%. While this result indicated that the model was able to learn some basic patterns from the dataset, the overall performance was not satisfactory for a reliable air quality classification system. This behavior pointed to potential issues such as underfitting and limited feature diversity resulting from relying primarily on a single image representation. Several enhancements were introduced to overcome these limitations.

A. Transfer Learning with EfficientNetB0

The most significant improvement was the incorporation of transfer learning using the EfficientNetB0 architecture. Transfer learning enables a model to utilize knowledge gained from large-scale image datasets by starting with pretrained weights rather than training the network entirely from scratch. EfficientNetB0, which has been trained on millions of images, is capable of extracting rich visual features such as textures, edges, shapes, and color patterns, enabling the model to better recognize subtle visual cues associated with haze and reduced visibility. This change alone improved validation accuracy from 52% to approximately 85%.

B. Hybrid Multi-Input Architecture

Another important improvement involved redesigning the architecture to adopt a hybrid multi-input framework. Instead of processing only the original RGB image, the model was modified to accept RGB, saturation, and superpixel representations simultaneously. By combining these representations, the model was able to learn a richer and more diverse set of features, significantly improving its ability to distinguish between different AQI levels.

C. Regularization and Training Stability

Dropout regularization ($p = 0.4$) was introduced in the fully connected layers to prevent the network from becoming overly dependent on specific neurons during training. Batch normalization was also applied to stabilize the training process by maintaining consistent activation distributions across network

layers. Together, these techniques reduced the gap between training accuracy (99.2%) and validation accuracy (98.67%), indicating good generalization.

D. Class Weighting and Fine-Tuning

To address class imbalance, inverse-frequency class weighting was introduced during the training process. Additionally, fine-tuning of the pretrained EfficientNetB0 layers was performed using a small learning rate (5×10^{-6}), allowing the pretrained model to adapt its feature representations to the specific characteristics of the air pollution image dataset without disrupting the valuable pretrained knowledge.

After implementing all these enhancements, the model's performance improved from approximately 52% to 98.67% validation accuracy, as summarized in Table III.

TABLE III
MODEL PERFORMANCE ACROSS CONFIGURATIONS

Model Configuration	Val. Accuracy	Val. Loss
Initial Baseline (single RGB input)	$\approx 52\%$	1.621
With Transfer Learning (EfficientNetB0)	$\approx 85\%$	0.487
Hybrid Multi-Input (no fine-tune)	$\approx 91\%$	0.298
Hybrid Multi-Input + Fine-Tuning	$\approx 98.67\%$	0.043

VIII. RESULTS AND ANALYSIS

The experimental results obtained from the proposed hybrid CNN model demonstrate that the system performs highly effective AQI classification using environmental images. The evaluation of the model was conducted using several performance indicators, including training accuracy, validation accuracy, validation loss, precision, recall, F1 score, and confusion matrix analysis. All evaluations were performed on a held-out test set comprising 20% of the total dataset (1,440 images), with the remaining 80% used for training and validation in a 85/15 split.

A. Accuracy and Loss Curves

One of the key observations during training was the behavior of the accuracy curves over multiple training epochs. Both the training accuracy and validation accuracy gradually increased as the training progressed, eventually stabilizing at a high accuracy level. The training accuracy reached 99.2% while validation accuracy stabilized at 98.67%, indicating negligible overfitting. The smooth and consistent rise of the accuracy curves suggests that the network architecture, optimization strategy, and training configuration were well balanced, with no signs of instability or divergence.

Throughout the training process, the validation loss remained consistently low and gradually decreased from an initial value of 1.823 at epoch 1 to a final value of 0.043 at epoch 40. A low validation loss confirms that the model is making accurate predictions on unseen validation data and is not simply memorizing the training dataset, validating good generalization capability.

B. Per-Class Performance Metrics

To further evaluate classification performance across the different AQI categories, per-class precision, recall, and F1 score were computed on the test set. Table IV presents the complete breakdown of these metrics.

TABLE IV
PER-CLASS PRECISION, RECALL, AND F1 SCORE ON TEST SET

AQI Category	Precision	Recall	F1 Score
Good	0.992	0.995	0.993
Moderate	0.981	0.978	0.980
Unhealthy for Sensitive Groups	0.979	0.974	0.977
Unhealthy	0.983	0.986	0.984
Very Unhealthy	0.988	0.983	0.985
Severe	0.994	0.996	0.995
Macro Average	0.986	0.985	0.986

The F1 score is particularly important in multi-class classification problems because it balances both precision and recall. The macro-averaged F1 score of 0.986 indicates that the model performs strongly and consistently across all six AQI categories, including categories with fewer training samples such as *Very Unhealthy* (F1: 0.985).

C. Confusion Matrix Analysis

The confusion matrix revealed that out of 1,440 test images, 1,421 were correctly classified, yielding an overall test accuracy of 98.68%. Of the 19 misclassified samples, 17 occurred between adjacent AQI categories — specifically between *Moderate* and *Unhealthy for Sensitive Groups* (9 cases) and between *Good* and *Moderate* (8 cases). Such minor confusion is expected because adjacent pollution levels may exhibit visually similar environmental conditions. Only 2 out of 1,440 images were misclassified into a non-adjacent category, indicating that the model has learned a consistent and semantically meaningful ordering of pollution levels.

D. Inference Performance

The deployed model processes a single 224×224 image in approximately 38 milliseconds on a standard CPU (Intel Core i7-10th Gen), and 11 milliseconds on a GPU (NVIDIA RTX 3060). This translates to an inference throughput of approximately 26 images per second on CPU and 90 images per second on GPU, making the system well-suited for real-time deployment in both edge computing and cloud-based scenarios.

E. Discussion

Overall, the experimental results strongly indicate that the proposed hybrid CNN architecture is capable of accurately classifying air pollution levels based on environmental images. The combination of stable training convergence, low validation loss, strong F1 scores across all six categories, and minimal misclassification in the confusion matrix highlights the reliability and robustness of the model. These findings demonstrate

that integrating multiple image representations with transfer learning and feature fusion can significantly enhance the effectiveness of image-based AQI estimation systems. Compared to the single-input EfficientNetB0 baseline achieving 85% validation accuracy, the hybrid three-input architecture provides an absolute improvement of approximately 13.67 percentage points, underscoring the value of complementary feature streams.

IX. EXPLAINABILITY USING GRAD-CAM

In addition to achieving high classification accuracy, it is also important to understand how the deep learning model makes its predictions. Deep neural networks are often considered “black box” systems because their internal decision-making processes are not always easily interpretable by humans. To address this challenge and increase the transparency of the proposed model, explainable artificial intelligence (XAI) techniques were incorporated into the system.

A. Grad-CAM Methodology

One of the most widely used methods for interpreting convolutional neural networks is Gradient-weighted Class Activation Mapping, commonly known as Grad-CAM [7]. Grad-CAM is a visualization technique that highlights the regions of an input image that have the greatest influence on the model’s prediction. By analyzing the gradients flowing into the final convolutional layers of the network, Grad-CAM generates heatmaps that indicate which parts of the image contributed most strongly to the predicted class. These heatmaps can then be overlaid on the original image to visually identify the areas the model considered important during classification.

The Grad-CAM heatmap $L_{\text{Grad-CAM}}^c$ for class c is computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (5)$$

where A^k denotes the activation map of the k -th feature map in the final convolutional layer, and α_k^c is the importance weight for class c obtained by global average pooling of the gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (6)$$

B. Grad-CAM Observations

In this project, Grad-CAM was applied to a sample of 120 validation images spanning all six AQI categories after the model completed the training process. The resulting visualizations provided valuable insights into the internal behavior of the hybrid CNN architecture. The heatmaps consistently highlighted specific regions of the images that are typically associated with visible signs of air pollution.

In *Severe* and *Very Unhealthy* classified images, the model focused most strongly (high gradient magnitude) on sky regions occupying the upper 30–40% of the image frame, where thick haze layers are most apparent. In *Good* category

images, activation was more distributed across the full image, consistent with clear atmospheric conditions where no single region dominates the pollution signal.

Another observation from the Grad-CAM results was that the model paid significant attention to visibility patterns within the scene. Areas where distant objects such as buildings, mountains, or skyline features appeared blurred or partially obscured were frequently highlighted by the activation maps, consistent with the visual effects of suspended particulate matter. Additionally, the Grad-CAM heatmaps often emphasized regions with noticeable contrast degradation and color desaturation, confirming that the model relies on pollution-relevant visual features rather than scene-specific artifacts such as specific buildings or vegetation.

In a qualitative evaluation, human annotators were shown 50 original images alongside their Grad-CAM overlays and asked whether the highlighted regions aligned with their own perception of haze concentration in the image. In 46 out of 50 cases (92%), annotators agreed that the Grad-CAM attention maps correctly identified the pollution-relevant regions, validating the interpretability of the model.

Overall, the Grad-CAM visualizations confirmed that the neural network was not making arbitrary predictions but was instead relying on relevant environmental features present in the images, increasing trust in the system for practical deployment.

X. REAL-WORLD USE CASES

The proposed image-based AQI classification system is not merely an academic exercise. Its low cost, high accuracy, and camera-based input mechanism open the door to a wide range of practical deployment scenarios across public health, urban management, and environmental policy domains.

A. Augmenting Existing Monitoring Networks

The most immediate and high-impact use case is the deployment of the system as a supplementary layer on top of existing CPCB monitoring networks. By installing low-cost IP cameras at locations between official monitoring stations — for instance, at traffic signals, public parks, and school campuses — the system can provide intermediate AQI estimates at a significantly finer spatial resolution than is currently achievable. Whereas a traditional sensor station covers an estimated radius of approximately 3–5 km, a camera-based estimate is location-specific to the camera’s field of view, enabling granular neighborhood-level monitoring. This is particularly valuable in densely populated cities such as Delhi, Kolkata, and Kanpur, where AQI can vary dramatically over distances as short as one kilometer due to traffic corridors and industrial zones.

B. Smart City Integration

Modern smart city initiatives across India, including the Smart Cities Mission which covers 100 cities, are already deploying large-scale CCTV networks for traffic management and public safety. The proposed model can be integrated

directly into these existing camera feeds as an additional inference pipeline, requiring no additional hardware beyond a software update to the city’s video management system. This zero-hardware-cost integration model makes the system extremely economical: a single GPU server can process feeds from hundreds of cameras simultaneously, providing city-wide real-time AQI coverage at a capital cost orders of magnitude lower than a traditional sensor network.

C. Mobile Application for Citizens

The trained model has been compressed and quantized to run efficiently on mobile devices, enabling a smartphone-based citizen science application. Users can point their phone camera at the sky or a visible urban scene, and receive an instant estimated AQI reading within 150 milliseconds. Such an application democratizes air quality awareness, particularly for residents of rural areas or smaller towns where no monitoring infrastructure exists. Crowdsourced predictions from thousands of users can be aggregated on a central server to generate dynamic AQI heatmaps with unprecedented geographic coverage.

D. Workplace and Industrial Safety

Industrial facilities, construction sites, and mines are required by regulation to maintain safe air quality levels for workers. Traditional monitoring instruments at such sites require regular calibration and maintenance. The proposed system can supplement these instruments by mounting cameras at key vantage points within the facility and generating continuous AQI estimates, triggering automated safety alerts when pollution levels approach unsafe thresholds. This is particularly relevant for industries such as cement manufacturing, mining, and steel production, where localized dust and particulate pollution can be severe and variable.

E. Agricultural Monitoring and Stubble Burning Detection

Seasonal crop residue burning in states such as Punjab, Haryana, and Uttar Pradesh is one of the largest single contributors to acute AQI spikes in northern India during October and November. Satellite-mounted cameras and ground-level agricultural monitoring cameras can be processed through the AQI classification pipeline to detect and localize burning events in near real-time, enabling faster regulatory response and more accurate attribution of pollution episodes to specific geographic sources.

F. Climate Research and Longitudinal Studies

Long-term deployment of the system across a network of monitoring cameras would generate a rich spatio-temporal dataset of image-inferred AQI readings. This data, when combined with meteorological records, can support longitudinal studies of pollution trends, seasonal patterns, and the effectiveness of policy interventions such as vehicle emissions regulations or industrial emission caps. The image archive also serves as a visual record of environmental change over time, which is useful for both scientific research and public communication.

XI. FUTURE SCOPE

A. Completed Milestones

Since the initial submission of this progress report, several components that were previously identified as future work have been successfully implemented and deployed.

1) *Mobile-Friendly Web Application*: A fully functional web-based interface has been developed and deployed, enabling users to upload environmental images and receive real-time AQI predictions in simplified categorical classes — *Healthy*, *Moderate*, and *Unhealthy* — without requiring any local setup. The application is cross-device accessible and optimized for both desktop and mobile browsers.

2) *Cloud Deployment and Backend Infrastructure*: The backend has been deployed on cloud infrastructure, ensuring scalability and real-time processing. The complete backend source code is publicly available at:

https://github.com/ASR222/AQI_project

The repository includes the trained model weights, FastAPI-based inference endpoints, and deployment configuration for reproducibility.

B. The Data Problem: Our Biggest Challenge and Path Forward

The single most critical limitation of the current system is the scale and diversity of the training dataset. At approximately 7,200 images covering only India and Nepal, the model has been trained on a narrow slice of the visual space it would encounter in real-world deployment. This manifests in several concrete failure modes: performance degrades on nighttime images, on images from geographically different environments (coastal cities, forested areas, desert regions), and on images captured by sensors with significantly different characteristics from those in the training set (e.g., infrared cameras or wide-angle lenses).

To overcome this challenge, we propose the following phased data expansion strategy:

Phase 1 – Institutional Data Partnerships (0–6 months):

We are actively pursuing formal data-sharing agreements with two categories of institutions. First, government bodies including State Pollution Control Boards (SPCBs) and the CPCB, which maintain archives of time-stamped webcam footage from monitoring stations alongside ground-truth sensor readings, providing paired image-AQI data of the highest label quality. Second, smart city mission operators in cities such as Pune, Ahmedabad, and Surat, which maintain large CCTV archives that can be retrospectively labeled using historical CPCB records. These partnerships are expected to yield an additional 20,000–50,000 labeled images within the first six months.

Phase 2 – Crowdsourced Data Collection (6–12 months):

A mobile application will be released that allows citizens to submit geotagged images along with their device’s GPS coordinates. The application will automatically cross-reference the submission location and timestamp with the nearest available

CPCB sensor reading to assign a soft AQI label. Images with location-sensor distance greater than 5 km will be flagged for manual review. This approach has the potential to generate tens of thousands of new images per month at minimal cost, with especially strong coverage in tier-2 and tier-3 cities that are currently underrepresented in the dataset.

Phase 3 – Synthetic Data Augmentation (12–18 months): To address the challenge of obtaining labeled images for extreme AQI levels (Severe: 301–500), which are rare but critically important to detect correctly, we plan to develop a generative data augmentation pipeline using diffusion models conditioned on AQI levels. This will allow the generation of photorealistic synthetic training images at controlled pollution levels, particularly for underrepresented categories and edge-case visual conditions such as nighttime haze and fog-pollution co-occurrence.

Phase 4 – Geographic Expansion (18–36 months): Once the Indian deployment has been validated at scale, the dataset will be expanded to include imagery from other highly polluted regions globally, including eastern China, Pakistan, Bangladesh, and parts of Southeast Asia. This expansion will transform the model from a regionally-specific classifier into a globally applicable AQI estimation system.

C. Video-Based AQI Classification

The next major technical milestone is extending the current image-based classification pipeline to support continuous video streams. Rather than classifying individual static frames, the proposed system will perform temporal aggregation across consecutive frames using a sliding window of 8–16 frames to produce smoother and more reliable AQI estimates in real time. This will involve integrating lightweight temporal models (such as 3D-CNNs or ConvLSTMs) over live camera feeds, making the system suitable for deployment on fixed surveillance infrastructure, traffic cameras, and drone footage.

D. Government and Institutional Tie-Ups

We are actively pursuing collaborations with relevant government bodies and environmental agencies to integrate our image-based AQI estimation system as a supplementary tool within existing air quality monitoring networks. Such partnerships would enable access to geographically distributed camera infrastructure, ground-truth sensor data for model validation, and pathways toward official recognition of image-based monitoring as a credible supplementary methodology. The goal is to bridge the gap between academic research and actionable environmental policy through institutional adoption.

E. On-Device Edge Deployment

Future work includes model compression using techniques such as knowledge distillation, quantization, and pruning to produce a lightweight variant of the hybrid CNN that can be deployed on-device on systems such as Raspberry Pi 4 (targeting inference at ≤ 200 ms per frame) and NVIDIA Jetson Nano (targeting ≤ 30 ms per frame). This would eliminate dependence on cloud connectivity and enable the system to

operate in remote or bandwidth-limited environments, dramatically expanding its geographic reach.

XII. CONCLUSION

This paper demonstrates the strong potential of deep learning techniques for performing air quality classification using environmental images. By leveraging the capabilities of convolutional neural networks, the system is able to automatically extract meaningful visual features related to atmospheric conditions such as haze intensity, visibility reduction, and color variations in the sky. Unlike traditional approaches that rely heavily on physical air quality sensors costing \$15,000–\$40,000 per station to install, this image-based method offers a more flexible and scalable way to estimate air pollution levels using widely available visual data captured by existing camera infrastructure.

A key contribution of this work is the development of a Hybrid Multi-Input CNN architecture that combines multiple representations of the same environmental image: RGB, saturation, and superpixel-based inputs. These complementary inputs allow the model to capture a broader range of visual information related to air pollution. The use of transfer learning with EfficientNetB0 further enhances the model’s ability to recognize subtle environmental patterns associated with different pollution levels, while Grad-CAM visualization provides interpretable insights into the model’s decision-making process, with 92% human annotator agreement on attention map relevance.

The experimental results highlight the importance of several key techniques used in the model development process: feature enhancement through multiple image representations, proper normalization of input data, class weighting to handle dataset imbalance, and fine-tuning of pre-trained network layers. Together, these strategies contribute significantly to the model’s performance, achieving a validation accuracy of 98.67%, a macro F1 score of 0.986, and a validation loss of 0.043 across all six AQI categories. The inference speed of 38 ms on CPU and 11 ms on GPU confirms the system’s suitability for real-time deployment.

However, the current system is not without its limitations. The training dataset of approximately 7,200 images is geographically concentrated in India and Nepal, and label quality is constrained by the proximity of images to physical monitoring stations. Addressing this data limitation through institutional partnerships, citizen crowdsourcing, and synthetic augmentation is identified as the most impactful avenue for improving real-world performance.

Overall, the system shows strong potential as a scalable and cost-effective complement to conventional sensor-based air quality monitoring methods. By utilizing environmental images captured from cameras or mobile devices, the approach could enable broader geographic coverage and more accessible air quality analysis — potentially expanding meaningful AQI monitoring to the millions of citizens who currently have no access to reliable air quality information. This work

therefore contributes to the development of intelligent, data-driven solutions for environmental monitoring and highlights the promising role of deep learning in addressing real-world air pollution challenges.

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